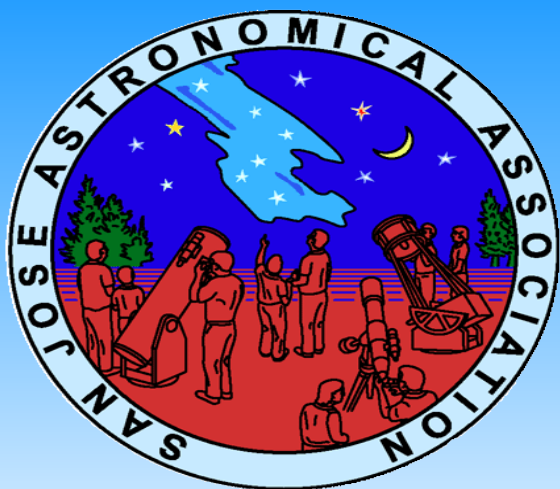




EPHEMERIS

November 2012

SJAA Activities Calendar

Jim Van Nuland

November

- | | |
|---|---|
| <p>4 DST ends at 2 a.m. Set clocks back one hour.</p> <p>4 Observing H-alpha flares and sunspots at Houge Park. Also our Telescope Tune-up Time. Sun party and tune-up hours: 2:00 until 4:00 p.m.</p> <p>9 Astronomy Class at Houge Park. 7:00 p.m. The topic: Amateur telescope making, why, how and where.</p> <p>9 Houge Park star party. Sunset 5:02 p.m, 19% moon rises 2:58 a.m. Star party hours: 7:00 until 10:00 p.m.</p> <p>10 Dark-Sky weekend. Sunset 5:01 p.m, 10% moon rises 4:07 a.m.</p> <p>17 Dark-Sky weekend. Sunset 4:56 p.m, 24% moon sets 9:12 p.m. Henry Coe Park's "Astronomy" lot has been reserved.</p> <p>23 Houge Park star party. Sunset 4:53 p.m, 83% moon sets 3:22 a.m. Star party hours: 7:00 until 10:00 p.m.</p> <p>25 Astronomical Swap Meet. Noon to late afternoon.</p> | <p>Meeting at 8:00. Our speaker is Peter Platzer discussing ArduSat - Making Space Exploration Affordable for everyone.</p> <p>2 Observing H-alpha flares and sunspots at Houge Park. Also our Telescope Tune-up Time. Sun party and tune-up hours: 2:00 until 4:00 p.m.</p> <p>7 Astronomy Class at Houge Park. 7:00 p.m. The topic: The Sun, planets and minor solar system objects.</p> <p>7 Houge Park star party. Sunset 4:50 p.m, 34% moon rises 1:46 a.m. Star party hours: 7:00 until 10:00 p.m.</p> <p>8 Dark-Sky weekend. Sunset 4:50 p.m, 24% moon rises 2:53 a.m.</p> <p>15 Dark-Sky weekend. Sunset 4:52 p.m, 11% moon sets 7:59 p.m. Henry Coe Park's "Astronomy" lot has been reserved.</p> <p>17 Imaging SIG meeting - Image Acquisition Software. Houge Park. 7:30 until 9:00 p.m.</p> <p>21 Houge Park star party. Sunset 4:54 p.m, 69% moon sets 2:12 a.m. Star party hours: 7:00 until 10:00 p.m.</p> <p>29 No meeting. Moved to Dec. 1</p> |
|---|---|

December

- 1 **General Meeting.** Board meeting at 6:00; Social Time at 7:30; General

24 hour news and information hotline:
(408) 559-1221
<http://www.sjaa.net>

Imaging Special Interest

Group

Ed Wong

The SJAA Imaging SIG held our third meeting on Sat. Oct 6. After 2 meetings at Houge Park in the Hall we met at the tennis courts at Houge Park to set up scopes and cameras to do some imaging. We had about 25 people come out for the meeting with about 14 bringing scopes and cameras. We did a complete walk through session from start to finish of setting up a mount to taking images. The areas we covered were polar alignment, setting up and balancing the scope & camera. We showed how to do a drift alignment of the mount, focusing with a Bahtinov mask, setting up and running an auto guider with PHD auto guiding software and finally we took some images of the Andromeda Galaxy.

The session was well received and attendees shared that they had learned a lot for the session and were motivated to try out some of the things they had learned from the session. We are planning the next SIG meeting with a few topics on using camera control and image acquisition software packages. The dates for the meetings will be announced very soon and posted on the SJAA main website. (See photo on page 6).

A comet, a dim eclipse, and some red blueberries

Akkana Peck

The news sites have been jumping with news of a newly discovered comet, C/2012 S1 (ISON), which could provide quite a show a year from now.

It's still in the outer solar system now, a bit more distant than Jupiter, but its perihelion, on November 28, 2013, will be unusually close to the sun, less than 2 million km. Compare that with Mercury, at almost 58 million km — this comet really is a sun-grazer. And it seems to be relatively large, possibly 3 km in diameter, so it might survive a close sun pass fairly well.

This has led to all sorts of wild predictions — it'll be brighter than the full moon! it'll be visible in the daytime! it'll be the brightest comet in human history! Don't believe any of it. Comets are notoriously unpredictable, and no one really knows what a comet will do, especially more than a year ahead of time. However, chances are that we'll get a nice show, a relatively bright, perhaps naked-eye comet gracing our skies. In the weeks leading up to perihelion, it will be a morning twilight object in Virgo and Libra.

The comet's name, ISON, refers to the International Scientific Optical Network. Two astronomers spotted it from Russia in late September using a 16" f/3 telescope. It's not much to look at now, 17th magnitude, so it's probably out of reach of most visual observers for a little while yet. (see http://www.astronomy.com/~link.aspx?_id=070aa63e-9b88-4893-afdb-db7e8e859730)

So what is in reach THIS month for more modest telescopes?

Jupiter, rising in mid-evening, makes two close passes with the moon, on November 1 and 28. In some parts of the world the moon will occult Jupiter, but here in California they only get within about a degree and a half of each other. Of course Jupiter is always a great

view whether or not it's near the moon, especially when it's close to opposition, like it is now.

Uranus and Neptune are visible all evening. Uranus is in a star-poor area of Pisces, with few pointer stars to help in finding it. Good thing it's usually an easy target.

Neptune is farther west, in Aquarius, also fairly far from any bright stars, but you can use the left horn of Capricornus to start your sweep.

Pluto isn't a good target this month, barely peeking into the predawn sky.

You can still catch Mars, barely, in the early evening. But it's very small — less than 5 arcseconds. Mars was in the news last month too: the Opportunity rover, though upstaged by it's newer, larger sibling Curiosity, is still doing great work. Opportunity snapped a photo of a rock covered with "spherules", similar to the Martian spherules previously called "blueberries" for their size and shape (but not color). But these are different from previous blueberries seen on Mars, because they have less iron and were probably formed by a different mechanism. (see http://science.nasa.gov/science-news/science-at-nasa/2012/14sep_myster)

Some of these new blueberries have been partially eroded away by wind, and you can see that the insides were hollow — as Opportunity's principle investigator said, the spherules are "crunchy on the outside, and softer in the middle".

When I first saw the photo, I immediately thought of the concretions I've seen in sandstone in the desert southwest, around Page, AZ and Kanab, UT. (see <http://shallowsky.com/geology/blueberries/>) But someone pointed out an even closer match. Go to youtube and search for the Invaders

from Mars (1953) Trailer. Pause it at 1:31, where they're showing the effects of the diabolical Martian melting ray. (<http://www.youtube.com/watch?v=Ury5b-qt1Y>) Then click back and forth between the two images. You'll be amazed! I don't know why NASA hasn't considered this explanation for the blueberries/spherules yet.

Mercury stays low this month: it's an evening at the beginning of November, then reappears in the morning sky at the end of the month. Saturn and Venus are morning objects all month.

And on November 28, early risers can catch a penumbral lunar eclipse. In a penumbral eclipse, the moon never quite makes it into the full shadow of the earth; but it gets deep enough in this eclipse in that you should be able to see the darkened parts of the moon fairly easily. The eclipse starts at 4:15 am and continues until moonset at 7:03, but the part before around 6am may be subtle and hard to notice.

Daylight Savings Time ends on November 4, so don't forget to set your clocks back! Assuming you're a luddite like me who still has old fashioned clocks that don't set themselves.

Directions to Houge Park

Houge (rhymes with "Yogi") Park is in San Jose, near Campbell and Los Gatos. From Hwy. 17, take the Camden Avenue exit. Go east 0.4 miles, and turn right at the light, onto Bascom Avenue. At the next light, turn left onto Woodard Road. At the first stop sign, turn right onto Twilight Drive. Go three blocks, cross Sunrise Drive, then turn left into the park.

From Hwy. 85, take the Bascom Avenue exit. Go north, and turn right at the first traffic light, onto White Oaks Road. At the first stop sign, turn left onto Twilight Drive. You will now be passing the park. Turn right at the first driveway, into the parking lot.

Messier Half-Marathon

Mark Wagner

On October 13, the San Jose Astronomical Association was out in force, at Henry Coe State Park. When I joined the SJAA, in the mid-90's, there were two places the club went. Coe, and Fremont Peak. In the past decade, the club's presence has dwindled, down to a few people, unannounced, occasionally going to these places.

Last night was a special occasion. SJAA member Teuro Utsumi had proposed an idea he had, as a club activity - a Half Messier Marathon. Teruo realized there were many new SJAA members who were inexperienced, and might enjoy a doable project (the full Marathon is daunting, a half though is much more "realistic" for most people) in a group setting, with some experienced observers to help, at a darker site. All in all, a good twenty five or more of us piled into the overflow lot at Coe, recreating the fun atmosphere we used to enjoy there every month.

It worked wonderfully. Although the skies were muted by Coe standards, the fun was anything but. Before sunset, we had a pot-luck BBQ. It proved to

be the perfect ice-breaker for people to chat and make some new friends. As Teruo talked to us about how the marathon would work, the sun dipped out of sight behind him and the coast range mountains silhouetted stark black against the deep blue sky, with gold/orange rays spreading up and over us. Vega popped out overhead, and we were off to a great start. Temps were pleasant, there was the slightest of breeze. Conditions seemed great.

The newcomers to "dark skies" (meaning, away from Hogue Park or their own backyards) had scopes ranging from 72mm to 12.5". One was a goto. I had my 18", but barely got to use it, as I had volunteered to help others star hop to the targets, or to identify them.

A funny thing happened to me. I'd get called over to "confirm" things like M23, M18, this glob or that... and the fact is, I hadn't looked at the Messiers in such a long time, I didn't recognize them! The only way I could confirm many of them was by looking at a chart (in this case, mostly on people's phones or iPads), and seeing where their scope was pointed. I

suppose there's some newbie left in me too.

While the transparency was down quite a bit... there was a lot of moisture in the atmosphere exacerbating San Jose's light dome, and probably some high thin cloud moving through, it was fine for catching the Messiers. People got a taste of what deep sky objects can look like, they learned that it is a short pleasant drive to a dark(er) site, they made new friends, and found success and enjoyment using their telescopes.

The mood all night was fun and festive, and I'm certain all would agree it was a great success. Kudos to Teruo for the initiative and work in putting this all together... well done!

Next spring, we should have the second half of the "Half Messier Marathon". I hope these sorts of events will make Coe what it used to be for the SJAA - an active observing site. I'd certainly be up for monthly 3rd Q Saturdays there. I have wonderful memories from years gone by, at Coe. Time to make some new ones.



On the left, Lee Hoglan is setting up his binoculars before starting the Messier Half-Marathon on October 13th at Coe Park. On the right, Mark Wagner is all ready to go with his 18" Dobsonian. Photos by Ed Wong.



A Cosmic Tease: Trials of the Herschel Space Telescope Science Teams

Dr. Marc J. Kuchner

Vast fields of marble-sized chunks of ice and rock spun slowly in the darkness this week, and I sat in the back of a grey conference room with white plastic tables spread with papers and laptops. I was sitting in on a meeting of an international team of astronomers gathered to analyze data from the Herschel Infrared Observatory. This telescope, sometimes just called Herschel, orbits the Sun about a million miles from the Earth.

The meeting began with dinner at Karl's house. Karl charred chorizo on the backyard grill while the airplanes

dribbled into Dulles airport. Our colleagues arrived, jetlagged and yawning, from Germany, Sweden, and Spain, and we sat on Karl's couches catching up on the latest gossip. The unemployment level in Spain is about twenty percent, so research funding there is hard to come by these days. That's not nice to hear. But it cheered us up to be with old friends.

The meeting commenced the next morning, as the vast fields of ice and rock continued to spin—shards glinting in the starlight. Or maybe they didn't. Maybe they didn't exist at all.

from the vicinity of these stars, like an infrared security camera, in case there was something else to be found lurking nearby.

And lo and behold, for a handful of stars, the bolometer measurements were off the charts! Maybe something was orbiting these stars. From the details of the bolometer readings—which channels lit up and so on—you would guess that this stuff took the form of majestic fields or rings of icy and rocky particles. It would be a new kind of disk, a discovery worth writing home to Madrid about.



Samuel Pierpoint Langley, who developed the bolometer in 1878. His instrument detects a broad range of infrared wavelengths, sensitive to differences in temperature of one hundred-thousandth of a degree Celsius (0.00001 C). In 1961, Frank Low developed the germanium bolometer, which is hundreds of times more sensitive than previous detectors and capable of detecting far-infrared radiation.

You see, this team is looking at a series of images of stars taken by a device called a bolometer that is blind to ordinary starlight. Instead, the bolometer inside Herschel senses infrared light, a kind of light that we would probably refer to as heat if we could feel it. But the idea of pointing the bolometer at the stars was not to collect ordinary starlight. It was to measure heat coming

There are several teams of astronomers analyzing data from the Herschel Space Telescope. They call themselves by oddly inappropriate sounding acronyms: GASPS, DUNES, DEBRIS. For the time being, the scientists on these teams are the only ones with access to the Herschel data. But in January, all the data these teams are working on will suddenly be released to the public. So they are all under pressure to finish their work by then. The team whose meeting I was sitting in on would like to publish a paper about the new disks by then.

But it's not so simple. The stars that this team had measured were relatively nearby as stars go, less than a few hundred light years. But the universe is big, and full of galaxies of all kinds—a sea of galaxies starting from maybe a hundred thousand light years away, and stretching on and on. Maybe one of those background galaxies was lined up with each of the stars that had lit up

the bolometer—fooling us into thinking they were seeing disks around these stars.

The team argued and paced, and then broke for lunch. We marched to the cafeteria through the rain. Meanwhile, vast fields of marble-sized chunks of ice and rock spun slowly in the darkness. Or maybe they didn't.

What else did Herschel recently uncover? Find out at <http://spaceplace.nasa.gov/comet-ocean>.

Dr. Marc J. Kuchner is an astrophysicist at the Exoplanets and Stellar Astrophysics Laboratory at NASA's Goddard Space Flight Center. NASA's Astrophysics Division works on big questions about the origin and evolution

of the universe, galaxies, and planetary systems. Explore more at <http://www.science.nasa.gov/astrophysics/>.

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

The Last Month in Astronomy

OCT-12-2012 **Dragon Unloaded** Commander Suni Williams and Flight Engineer Aki Hoshide used the ISS's robotic arm to grab and connect the Dragon spacecraft. This craft took up 882 pounds of supplies and will return with more than twice the amount of stuff to be returned to Earth. Splashdown is scheduled for October 28. Dragon is the supply ferry launched by the private firm Space X. <http://spacefellowship.com/news/art30562/station-crew-unloading-dragon-conducting-science.html>

OCT-10-2012 **Dry Ice at Venus** Venus is the hottest planet in the solar system, even hotter than Mercury. But scientists have found that the temperature high in the Venusian atmosphere can be quite cold. In fact, it can be cold enough for carbon dioxide to freeze out as snow. It would be about 125 km high. <http://www.astrobio.net/pressrelease/5068/a-curiously-cold-layer-in-venus-atmosphere>

OCT-09-2012 **X-ray Nova** The Swift space-based X-ray telescope has discovered a very rare X-ray nova near the center of the Milky Way. This is probably a dark hole with a stellar mass. The object now named Swift J1745-26 hit a peak in hard (higher energy) X-rays on September 18 but continued to brighten after that in soft X-rays. The object had been detected in other wavelengths but the atmosphere blocks X-rays. Boris Sbarufatti from Milan said "Once the X-rays fade away, we hope to measure its mass and confirm its black hole status. <http://www.astronomy.com/en/News-Observing/News/2012/10/The%20Swift%20satellite%20discovers%20a%20new%20black%20hole%20in%20our%20galaxy.aspx>

OCT-04-2012 **Deep Impact: Act 3** The NASA spacecraft Deep Impact studied comet Tempel 1 by using an impactor and running through the debris field. This was done on July 4, 2005. The craft was then redirected to comet Hartley 2 and it flew by that comet on November 4, 2010. Now it's engines have been fired up again. This time the potential target is the near-Earth asteroid 2002 GT. The rocket fired for 71 seconds and it changed the speed of the craft by 4.5 mph. <http://www.jpl.nasa.gov/news/news.php?release=2012-313>

OCT-03-2012 **New accuracy for Hubble** The Hubble constant, the speed at which the universe is expanding, has been measured to a new level of accuracy thanks to measurements by the Spitzer telescope. 10 Cepheid variable stars in the Milky Way and 80 more in the Large Magellanic Cloud were studied. By using the Spitzer's Infrared capabilities these Cepheids could be studied without having to worry about dust. This helped to improve the accuracy of distance measurements that use this "standard candle". Lead author of this study is Wendy Freedman of the Observatories of the Carnegie Institution for Science in Pasadena. She said "Just over a decade ago, using the words 'precision' and 'cosmology' in the same sentence was not possible, and the size and age of the universe was not known to better than a factor of two. Now we are talking about accuracies of a few percent. It is quite extraordinary. <http://www.jpl.nasa.gov/news/news.php?release=2012-309>

SEP-24-2012 **Milky Way Halo** The Milky Way galaxy has a halo of hot (100,000 to 1,000,000 Kelvins) gas that has a mass that might be equal to the mass of the Milky Way galaxy itself. No, this doesn't say anything about dark matter but it might be a solution to the missing baryonic matter question. (See question #3 in the August 2012 issue of the Ephemeris under "Astro Questions"). <http://chandra.harvard.edu/photo/2012/halo/>

SEP-06-2012 **ARP 116** Astronomer Halton Arp created a catalog of galaxies called the "Atlas of Peculiar Galaxies". Hubble has recently released an image of Arp 116, actually a pair of galaxies. One galaxy, also known as M60, is a large elliptical galaxy. It has a "companion", the spiral galaxy 4647. The two galaxies look like they might be touching but so far they don't appear to be pulling on each other. Instead, the two galaxies present a textbook example of the difference between ellipticals and spirals. The spiral contains many more new stars and therefore looks quite a bit more blue than the elliptical which is not creating a lot of new stars at this time. <http://www.astronomy.com/en/News-Observing/News/2012/09/Odd%20galaxy%20couple%20on%20space%20voyage.aspx>

SJAA Activities

Here are some pictures from just a few recent SJAA activities. Clockwise from the photo on right: 1) Some solar viewing was made available at the Almaden Art&Wine festival in September. The two young girls shown were part of a dance troupe. They later pulled the entire company over to see the flares and prominences; 2) Rob Jaworski was at the same festival, shown here with his "Mountie" hat; 3) Steve Gottlieb was the speaker at the General Meeting on September 23, here he is making a point about M51; 4) Michael Packer is setting up the club's new solar telescope at the Art&Wine festival; 5) The Imaging Special Interest Group set up on the tennis courts at Houge Park on October 6; 6) Pumpkin ready for an eclipse.



Photos 1, 2, 4 and 5 by Ed Wong; Photo 3 by Paul Kohlmler; Photo 6 by Michael Packer



It Must Be Astronomical ...

“For me, I am driven by two main philosophies: know more today about the world than I knew yesterday and lessen the suffering of others. You’d be surprised how far that gets you.” - Neil deGrasse Tyson

Loaners

The telescope loaner program has been revamped. The program now includes QuickSTART, program geared to those new to astronomy. Please check it out at <http://www.sjaa.net/loaners.shtml>.

School Star Parties

Completed Events					
	Total Sched.	Good Sky	Partial Success	Cloudy Fail	Cancel at noon
Jul	2	2			
Aug	1	1			
Sep	1	0	1		
Oct	-	-			
Nov	-	-			
Dec	-	-			
Total	4	3	1		
Scheduled - for the 2012/2013 school year					
	Total	Firm	Working		
Oct	8	7	1		
Nov	8	7	1		
Dec	4	3	1		
Jan	6	4	2		
Feb	1	1	0		
Mar	2	1	1		
Total	29	23	6		

As of September 25, 2012

School Star Party Link

For information on school star parties including how to schedule one see <http://www.sjaa.net/school.shtml>.

Officers and Board of Directors

Pres Mark Wagner
VP Greg Claytor
Sec Rob Jaworski
Tres Michael Packer
Dir Lee Hoglan
Dir Rich Neuschaefer
Dir Rod Norden
Dir Kevin Roberts
Dir (Open)

School Star Party Chairman

Jim Van Nuland (408) 371-1307

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 Ephemeris ephemeris@sjaa.net

Other e-mail contacts are available at <http://www.sjaa.net/contacts.html>

Members Email Lists:
<http://www.sjaa.net/majordomo.html>

<http://sanjoseastronomy.blogspot.com/>
 twitter: [sj_astronomy](#)
<http://www.meetup.com/A-A-N-C/>

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San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

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☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

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Send e-mail to membership@sjaa.net

Bring this form to any SJAA Meeting or send to the club address (above). Please make checks payable to "SJAA".

You can join or renew online:
<http://www.sjaa.net/membership.shtml>

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